LumenCore Evidence Control Plane

Public-Safe Solution Brief Draft for the Sovereign Defense Cloud

Commercial Solutions Opening W912HZ26SC005
U.S. Army Engineer Research and Development Center
High Performance Computing Modernization Program

Primary scope: Unified Service Layer and Vendor Lock-In Prevention. Integration boundaries: AI-Powered Orchestration evidence and Secure Data Fabric metadata.

PUBLIC-SAFE DRAFT - REQUIRED PRICE AND PRIVATE SAM-MATCHED IDENTITY ARE NOT INCLUDED

Website: https://lumen-core.ai
Public repository: https://github.com/robertashworth1986-debug/lumen-core-public
Prepared July 29, 2026

Acronyms and Abbreviations

This list is separate from the five-page proposal body, consistent with the July 20, 2026 Frequently Asked Questions.

Acronym	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CAC	Common Access Card
CLI	Command-Line Interface
CPU	Central Processing Unit
CSO	Commercial Solutions Opening
DoD	Department of Defense
DSRC	DoD Supercomputing Resource Center
ERDC	Engineer Research and Development Center
GFE	Government Furnished Equipment
HPC	High Performance Computing
HPCMP	High Performance Computing Modernization Program
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
ML	Machine Learning
MOSA	Modular Open Systems Approach
OpenAPI	Open standard for describing HTTP application interfaces
ROM	Rough Order of Magnitude
SAM	System for Award Management
SDC	Sovereign Defense Cloud
SHA-256	Secure Hash Algorithm with a 256-bit digest
SLSA	Supply-chain Levels for Software Artifacts
Zero Trust	Security model that continuously verifies access decisions

All substantive pages that follow use the full term at first use where practical. Tables and diagrams are included in the five-page body count.

1. Mission Gap and Proposed Solution

The High Performance Computing Modernization Program must coordinate hybrid resources, workflows, policy, and data without binding mission decisions to one vendor. LumenCore proposes an evidence control plane that records what was requested, which policy and orchestration path was selected, what artifacts were produced, and whether the delivered offline verifier can reproduce the resulting receipt.

LumenCore is not proposed as a replacement for the five DoD Supercomputing Resource Centers, a complete cloud platform, a cross-domain solution, or a security authorization. It is a modular validation and observability component that can be integrated by the Government or a prime platform provider.

Mission effectiveness

- Give operators one evidence format across government-owned and commercial environments.
- Designed to flag policy, configuration, adapter, and artifact drift before a result is promoted.
- Preserve adverse outcomes and abstentions instead of reporting only successful runs.
- Allow reviewers to verify a receipt offline without access to the originating control plane.

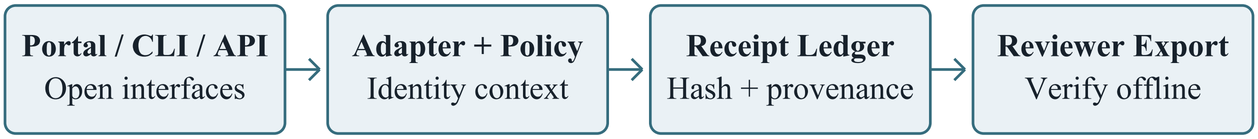
Focus-area alignment

Focus area	LumenCore contribution
Unified Service Layer	Primary: OpenAPI event and evidence interfaces spanning portal, command-line, and automation clients.
AI-Powered Orchestration	Integration boundary: policy-aware evidence hooks record orchestration decisions without replacing the selected scheduler.
Secure Data Fabric	Integration boundary: schema, provenance, tag, policy-result, and integrity receipts; workload payloads remain outside the default evidence path.
Vendor Lock-In Prevention	Primary: replaceable adapters, portable schemas, offline verification, and no mandatory proprietary cloud service.

Innovation

OpenTelemetry Logs Data Model 1.59.0 and SLSA Build Provenance 1.2 with in-toto Statement v1 are complementary interoperability contexts, not ranked competitors. The bounded LumenCore mechanism composes locally hash-linked event integrity, a hashed predeclared gate set, adverse-outcome retention, and offline verification at the workflow decision boundary. The local experiment scores only LumenCore full-control and ablation profiles. This is a mechanism distinction, not a superiority claim.

2. Modular Architecture and Data Boundary



Replaceable adapters: Kubernetes | OpenStack | scheduler | object store | observability

Component	Phase II behavior
Open adapter layer	Maps selected portal, command-line, scheduler, object-store, and observability events into versioned OpenAPI contracts and JSON event schemas.
Policy registry	Records policy identifiers, versions, inputs, outcomes, and exception reasons; it does not replace Government authorization services.
Receipt ledger	Builds locally hash-linked SHA-256 chains for request, decision, artifact, and verification metadata; the terminal root is supplied separately to the verifier.
Offline verifier	Checks schema, rehashes included artifact bytes, verifies chain continuity and declared rules, and reports missing evidence without network access.
Reviewer export	Produces bounded machine-readable and human-readable packets with limitations and failed checks retained.

Data minimization and sovereignty

The default evidence path stores control metadata, schema references, policy results, hashes, timestamps, and artifact locators. Mission data and model payloads remain in the authorized environment unless the Government defines a narrower approved data flow. Each classification level would use a separately deployed instance and enclave-approved interfaces with absolute data separation. This draft does not claim cross-domain transfer or classified-data certification.

Open replacement boundary

Adapters implement versioned contracts around Kubernetes, OpenStack, workload schedulers, storage, and observability systems using widely adopted protocols such as OpenAPI and Government-selected storage interfaces. A component or cloud may be replaced while the evidence contract, workload portability, and offline verifier remain stable. Government-selected identity and access services provide identity context; Common Access Card authentication is not assumed to be the sole path.

3. Phase II Prototype Plan and Feasibility

The following sixteen-week plan is a proposal assumption for Phase II prototype development, not an ERDC-promised schedule. Phase III demonstration and Phase IV implementation costs and activities are excluded from the required Phase II-only price basis.

Proposed period	Activity	Exit evidence
Weeks 1-3	Lock one unclassified use case, interfaces, acceptance rules, and boundary.	Approved interface and test protocol; no production connection.
Weeks 4-8	Build two adapters, policy registry, receipt chain, and offline verifier.	Schemas, tests, software bill of materials, and build receipt.
Weeks 9-12	Replay in advisory, human-in-the-loop shadow mode across two approved test environments.	Retained pass, fail, abstain, missing-data, override, and drift cases.
Weeks 13-16	Demonstrate cloud-agnostic portability, verification, phased handoff, and rollback.	Government-run verification, manual override, and limitation register.

Proposed acceptance protocol

The local precursor uses 48 deterministic synthetic workflows and seven tamper cases across one full-control and three ablation profiles, with a separate local anchor. OpenTelemetry 1.59.0 and SLSA 1.2/in-toto v1 are unranked context. This is not an HPCMP workload, external trust root, independent validation, or superiority evidence. Phase II would prelock the workflow, Government comparator, exclusions, window, and overhead budget. AI remains advisory with manual override.

Acceptance dimension	Proposed measurable check
Integrity	Detect all declared policy, deletion, ordering, digest, and gate mutations; any miss fails the controlled experiment.
Portability	One protocol and schema run through two replaceable environment adapters.
Failure visibility	Retain 100% of required fail, abstain, missing-input, policy-denied, and override cases; any omission fails.
Reproducibility	A Government reviewer reruns the delivered verifier from a clean environment; any receipt mismatch fails.
Cost efficiency	Against one selected baseline and fixed window, report bytes/event, capture and verify latency, storage/day, review minutes, and egress bytes.

Government inputs and assumptions

- One unclassified representative workflow, approved data and interfaces, identity context, endpoints, and security constraints.
- One approved comparator, two test environments, fixed window, and Government-controlled trust anchor or signing path.
- A Government reviewer approves the protocol and overhead budget and runs the verifier; no production access or GFE is assumed.

4. Security, Scalability, Operations, and Risk

Zero Trust integration boundary

LumenCore consumes identity, device, workload, policy, and environment context supplied by Government-approved services and records the decision evidence. It does not issue credentials, replace access-control authorities, or treat one authenticator as sufficient for every user. Each receipt can bind the applicable policy version, decision result, exception, and verifier outcome.

Classification and enclave pattern

The architecture can be considered for Unclassified, Secret, Top Secret, and caveated environments through separate enclave deployments and approved interfaces. Phase II should begin unclassified. This is an architectural pattern only; no classified handling, cross-domain transfer, Special Access Program support, or Sensitive Compartmented Information accreditation is claimed.

Scalability and cost control

- Partition receipts by enclave, mission, tenant, and retention policy without changing the verifier schema.
- Use content hashes and locators instead of duplicating large HPC artifacts in the evidence ledger.
- Report capture and verify latency, bytes per event, storage growth per day, operator-review minutes, and egress bytes over the same fixed window as the selected baseline.
- Keep adapters and storage replaceable so the Government can compare lifecycle cost and portability.
- Exercise both workload portability and bounded burst behavior without coupling the design to one cloud.

Primary risks and controls

Risk	Control
Undefined legacy interfaces	Use a versioned adapter contract and two bounded interfaces; isolate vendor-specific code.
Sensitive data exposure	Store metadata and hashes by default; keep payloads in enclave; approve expanded flows.
Performance overhead	Benchmark capture separately; permit asynchronous finalization where constraints require it.
Control or overhead failure	Stop and roll back on any declared attack miss, adverse-case omission, verifier mismatch, or Government-set overhead breach.
Security accreditation	Deliver review evidence; do not represent the prototype as authorized to operate.

Delivery, compute, and support boundary

The founder is the proposed technical lead for the evidence module and public code. The Government or selected prime owns authorized interfaces, access, security, and integration; an evaluator role is requested but no independent evaluator is committed. Surrogate development uses contractor-furnished commodity CPU, local storage, and open software; no HPC allocation or cloud capacity is claimed. Staffing, compute, support, and transition commitments must be bound in the private Phase II price and accepted boundary before work begins.

5. Commercial Readiness, Phase II Price Gate, and Evidence

Commercial approach

LumenCore is modular software plus integration and verification services using commercial technologies and open interfaces. It applies established interface, container, hashing, provenance, and observability patterns in an evidence-control role. The named profiles are unranked contexts. The SDC module is not proven; any resultant award is expected to be firm-fixed price.

Evidence	What it supports	What it does not support
Repository and local draft	Public patterns; the exact July 29 receipt remains local pending publication.	External reproducibility, deployment, production readiness, or field validation.
Offline verifier pattern	Offline receipt checks and explicit failure reporting.	Independent validation, Government acceptance, or classified accreditation.
Synthetic control ablation	48 deterministic synthetic workflows; full controls detected 7/7 declared tamper cases with complete adverse-case retention and synthetic artifact-byte rehash; each ablation lost a declared control relative to a separately pinned local anchor.	An external trust root, HPCMP performance, Government acceptance, independent validation, or superiority over the purpose-bounded standards.

Phase II Rough Order of Magnitude control

SUBMISSION BLOCKER: The CSO requires an estimated price for Phase II prototype development only. This public draft includes no price. Labor, infrastructure, support, indirect cost, profit, payment timing, and firm-fixed-price risk require review before one estimate is inserted privately.

Final submission gates

Required finalization	Current state
Phase II Rough Order of Magnitude	No price is included; a reviewed private estimate is required.
SAM identity and contact	Insert privately; verify exact match, all-awards status, and proposal email.
Portal and authority	Recheck Submittable, amendments, terms, and final confirmation.

Bounded next decision

If ERDC considers the module promising, the requested next step is a short technical clarification or solution pitch to lock one representative unclassified workflow, the integration boundary, Government-run acceptance checks, and the appropriate contract or agreement path.

Official project source: https://www.erdcwerx.org/sovereign-defense-cloud-for-high-performance-computing/

Funding is not currently available; this market-research lane does not guarantee an award.

Claim boundary: No ERDC selection, available funding, award, DoD deployment, authorization to operate, classified handling, field validation, customers, revenue, or realized savings, or performance beyond the bounded evidence is claimed.